

ORIGINAL RESEARCH

ATRIAL FIBRILLATION

Outcomes of Left Atrial Appendage Occlusion in Patients With Hypertrophic Cardiomyopathy, Rheumatic Heart Disease, and Cardiac Amyloidosis



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ABSTRACT

BACKGROUND Left atrial appendage occlusion (LAAO) has emerged as an effective stroke- prevention strategy for selected patients with nonvalvular atrial fibrillation (NVAf). However, LAAO outcomes data in patients with hypertrophic cardiomyopathy (HCM), rheumatic heart disease (RHD), or cardiac amyloidosis (CA), are limited.

OBJECTIVES This study aimed to compare the safety and efficacy of LAAO in patients with NVAf, with and without comorbid HCM, RHD, or CA.

METHODS Using OptumLabs Data Warehouse, a retrospective cohort of adult patients undergoing LAAO (2015-2023) was analyzed. Outcomes included mortality, stroke/transient ischemic attack (TIA), and bleeding, with multivariable Cox models and subgroup analyses.

RESULTS A total of 14,755 patients (mean age 76.5 ± 7.0 , 43.7% female, median follow-up 1.4 [0.8-2.4] years) were included. Compared with patients with AF, patients AF + RHD had high risk of nongastrointestinal/intracranial bleeding events (HR: 1.24; 95% CI: 1.04-1.49; $P = 0.02$), whereas AF + CA showed higher risk of composite endpoint (mortality, stroke/TIA, bleeding) (HR: 1.63; 95% CI: 1.17-2.27; $P = 0.004$), stroke/TIA (HR: 2.00; 95% CI: 1.13-3.54; $P = 0.02$), and gastrointestinal bleeding (HR: 2.50; 95% CI: 1.14-5.47; $P = 0.02$). There were no significant differences in clinical outcomes between patients with AF alone and those with AF + HCM.

CONCLUSIONS Patients with AF and either RHD or CA experienced higher bleeding rates following LAAO compared with those without these conditions, despite similar stroke/TIA rates in AF + RHD, suggesting a higher inherent bleeding risk and possibly further supporting a role of LAAO. Importantly, there was no difference in outcomes between patients with AF and HCM vs those without. Because of the small sample size, the results in HCM and CA cohorts are mainly hypothesis generating. (JACC Clin Electrophysiol. 2026;12:1089-1101) © 2026 by the American College of Cardiology Foundation.

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ABBREVIATIONS AND ACRONYMS

CA = cardiac amyloidosis

DOAC = direct oral
anticoagulants

HCM = hypertrophic
cardiomyopathy

LAAO = left atrial appendage
occlusion

NVAF = nonvalvular atrial
fibrillation

RHD = rheumatic heart disease

TIA = transient ischemic attack

VKA = vitamin K antagonists

Risk stratification for thromboembolism in nonvalvular atrial fibrillation (NVAF) is generally guided by clinical scoring systems, with anticoagulation typically recommended for those at elevated risk.¹ For patients with NVAF and contraindication to long-term anticoagulation therapy, left atrial appendage occlusion (LAAO) has emerged as an alternative strategy to reduce the risk of thromboembolic events.²⁻⁴

Although the efficacy and safety of LAAO has been well established in general NVAF populations deemed unsuitable for long-term anticoagulation therapy,²⁻⁵ safety and efficacy data remain limited regarding its use in special subgroups such as those with hypertrophic cardiomyopathy (HCM), rheumatic heart disease (RHD), and cardiac amyloidosis (CA).^{1,6} These conditions are associated with unique pathophysiological mechanisms and a higher prevalence of thrombus formation outside the left atrial appendage,^{7,8} which may influence long-term outcomes. Moreover, thromboembolic risk in these patients is elevated regardless of conventional scoring systems, and anticoagulation is typically recommended independent of this. Given the paucity of evidence, the present study aims to evaluate the clinical outcomes in patients with atrial fibrillation (AF) and to evaluate the clinical outcomes in patients with atrial fibrillation (AF) alone and AF with HCM, RHD, or CA undergoing LAAO, with a goal of informing risk-to-benefit considerations in these complex populations relative to the general population.

METHODS

STUDY POPULATION. This was a retrospective cohort analysis using deidentified administrative claims data from the OptumLabs Data Warehouse (OLDW), which includes medical and pharmacy claims and enrollment records for commercial and Medicare Advantage enrollees. The database contains longitudinal health information for more than 200 million enrollees and patients, representing a mixture of ages and geographic regions across the United States.^{9,10} The Mayo Clinic Institutional Review Board exempted this study from the review, as the study used preexisting, deidentified data.

The study population included adult patients aged 18 years or older who had undergone LAAO between March 13, 2015, and December 31, 2023, with preexisting diagnoses of AF. Patients were required to have at least 12 months of continuous medical and

pharmacy coverage (baseline period, single insurer) before the LAAO procedure (index date). Patients were assigned to mutually exclusive groups based on whether they had AF in addition to HCM, RHD or CA before LAAO ([Supplemental Table 1](#)) and compared with patients with AF without any of these conditions (group referred to as AF alone). Patients with AF and HCM, RHD, or CA are collectively referred to as the AF + SHD (structural heart disease) group. HCM and RHD were defined by 1 diagnosis for the respective condition in the 12 months before the LAAO. CA was defined as one diagnosis of amyloidosis and 1 diagnosis of heart failure or cardiomyopathy in the 12 months before the LAAO.

COVARIATES. Patient demographic characteristics included age, sex (female, male), race/ethnicity (non-White, White), and household income range. Patients were required to have at least 12 months of continuous medical and pharmacy coverage before LAAO. Comorbidities included coronary artery disease, hypertension, diabetes, venous thromboembolism, peripheral artery disease, anemia, chronic kidney disease, heart failure, liver disease, and alcoholism. Events (heart failure hospitalization, stroke, transient ischemic attack [TIA], bleed) and valve procedures in the previous 12 months were flagged. CHAD₂S₂-VASc and HAS-BLED scores were calculated for each patient but only the components were included in the models. In addition, cardiovascular medications (antiplatelets, nonsteroidal anti-inflammatory drugs [NSAIDs], warfarin) filled within 90 days before the LAAO procedure were captured.

OUTCOMES. Patients were followed until the end of study period (June 30, 2024), end of enrollment, or death, whichever occurred first. Mortality information withing the database was sourced from many different sources including the Social Security Administration Death Master File, medical charts, medical claims, disenrollment records and beneficiary reports. Outcomes included all-cause mortality, stroke/TIA (ischemic stroke, systemic embolism, TIA), bleeding (gastrointestinal, intracranial, other) in addition to a composite outcome (all-cause mortality, stroke/TIA, and bleeding). Other bleeding events are defined in [Supplemental Table 1](#). Mortality data are updated monthly and gathered from many different sources including Social Security Association Death Master File, medical charts, medical claims, disenrollment records and beneficiary report information. Stroke/TIA and bleed events were defined as a primary diagnosis on an emergency department visit or an inpatient admission date.

STATISTICAL ANALYSIS. Baseline patient characteristics were reported as frequencies (percentages) for categorical data and as means (standard deviation) for continuous variables. Unadjusted event rates or medians (IQR) were reported for each outcome using the number of patients with the given event as numerator and person-years (time to first event, death, or end of enrollment) as denominator. Cox proportional hazard regressions with multivariable adjustment were used to compare those with AF only with those with AF and HCM, RHD, or CA for each outcome. Covariates for multivariable adjustment were selected a priori based on clinical relevance and previous literature and included age, sex, race, income, comorbidity components of CHA₂DS₂-VASc and HAS-BLED, stroke/TIA/bleeding/heart failure hospitalization events or valve procedure in the previous 12 months, and concurrent medication use (antiplatelets, NSAIDs, warfarin). The Fine and Gray method was used to consider death as a competing risk when assessing nonfatal outcomes.¹¹ For the proportional hazard assumption (PH) we evaluated the comparisons between AF only and AF+SHD, AF+RHD/CA/HCM using a time-dependent covariate (interaction term of group [AF only or AF + SHD] and in [time]) for each outcome, with all *P* values > 0.05, suggesting no violation, except in bleeding for AF + SHD and AF + RHD. Finally, the residual confounding was assessed by testing 3 falsification endpoints that are unlikely to be a result of the LAAO procedure but might be related to unmeasured confounders such as frailty, pneumonia, fracture, and chronic obstructive pulmonary disease. Like the primary outcomes, the falsification endpoints were captured based on primary diagnosis on an emergency department visit or an inpatient admission date. Cumulative incidence curves were plotted to examine the outcomes at different time points.

SENSITIVITY AND SUBGROUP ANALYSES. Subgroup analyses included stratification by condition (HCM, RHD, CA), sex (male, female), age (<75 years, ≥75 years) and race (non-White, White) for each outcome and falsification endpoint.

A value of *P* < 0.05 was considered statistically significant for all tests. All tests were 2-sided. All analyses were conducted using SAS Enterprise Guide 7.1 (SAS Institute) and STATA 16.0 (STATA Corp).

RESULTS

STUDY COHORT. Of 22,083 enrollees with LAAO between March 2015 and December 2023, 14,755 met eligibility criteria and were included in the analysis

(Figure 1). Among these, 10,119 had AF alone, whereas the remaining 145 had AF + HCM, 60 AF + CA, and 4,431 patients had AF + RHD. The median follow-up was 1.4 (0.8–2.4) years.

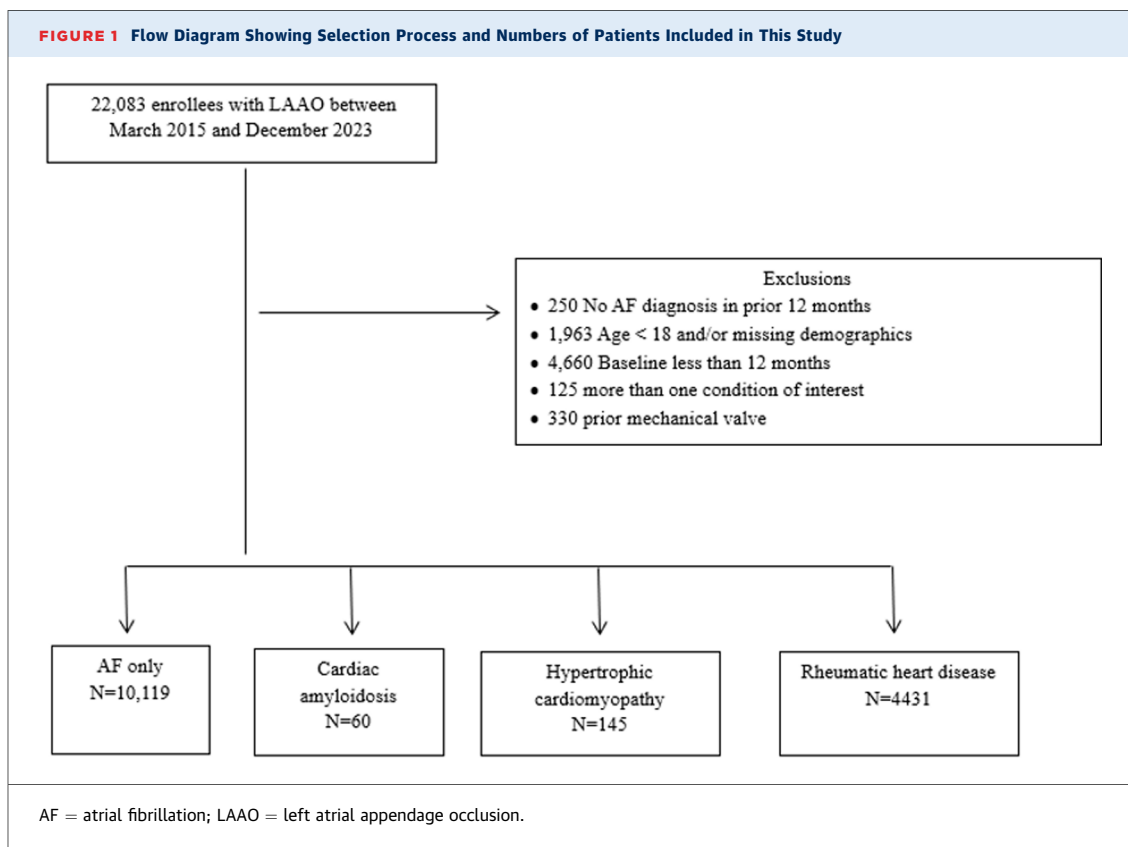
BASELINE CHARACTERISTICS. Table 1 summarizes the baseline characteristics of the study population. The mean age for patients with AF only was 76.1 ± 7.0 years, with minor but significant differences across the AF + SHD groups. All 4 cohorts had more male than female patients. The majority of patients were White (80.7%). Hypertension was the most prevalent (95.6%) comorbidity, followed by vascular disease (68.5%) and anemia (61.1%).

Both the CHAD₂S₂-VASc (mean 4.6 ± 1.4) and HAS-BLED (mean 3.0 ± 1.0) scores indicated a high-risk profile for many patients.

OVERALL OUTCOMES. Comparative analysis of outcomes between the patients with AF alone and those with AF + SHD is presented in Table 2 and Figure 2. The all-cause mortality rate was slightly higher in the AF + SHD group (12.09 per 100 person-years) compared with the AF-only group (9.48 per 100 person-years), (*P* = 0.22). Similarly, stroke/TIA rates were comparable between the 2 groups (4.76 vs 5.15 per 100 person-years), with a nonsignificant HR: 0.98 (95% CI: 0.87–1.11; *P* = 0.78). There was no significant difference in rates of gastrointestinal bleeding (HR: 1.05 [0.86, 1.27], *P* = 0.67) and intracranial bleeding (HR: 1.00 [0.90, 1.11], *P* = 0.96). However, a significant difference was observed in the rate of other bleeding events, with an HR: 1.25 (95% CI: 1.05–1.49, *P* = 0.01) for the AF-SHD group.

The timing of bleeding events was comparable between the AF and AF + SHD groups (22.4% vs 27.1% within 30 days, *P* = 0.01 and up to 74.3% vs 73.7% within 1 year, *P* = 0.78; Supplemental Table 2). Before LAAO, the most common anticoagulants used were direct oral anticoagulants (DOACs)—82.0% in AF-RHD and up to 93.8% in AF-CA—followed by vitamin K antagonists (VKAs): 6.3% in AF-CA and up to 18.5% in AF-RHD. Post-LAAO, the same pattern of anticoagulant use was observed; DOACs were the most common (85.3% in AF-RHD and up to 91.1% in AF-CA), followed by VKAs (8.9% in AF-CA and up to 21.7% in AF-RHD).

Length of anticoagulation use post-LAAO was comparable at 30 days with no statistically significant results (*P* > 0.05) when AF only was compared with each of the 3 SHD cohorts. However, at 3 months, 16.4% of the AF-only cohort was on anticoagulation as opposed to 18.7% of patients with AF + HCM (*P* = 0.46), 16.0% patients with AF + RHD (*P* = 0.51), and 29.8% of patients with AF + CA (*P* = 0.006); Supplemental Table 3.



AF + HCM. All-cause mortality rates were similar between the AF alone and AF + HCM group (9.48 per 100 person-years for AF alone and 8.20 per 100 person-years for AF with HCM), yielding an HR: 0.88 (0.57, 1.35); $P = 0.55$ (Figure 3). Similarly, the association with the composite outcome, stroke/TIA and bleeding risk was not significant ($P \geq 0.05$) (Table 3, Central Illustration).

AF + RHD. The AF + RHD group was the largest of the subgroups analyzed in this study. All-cause mortality was slightly higher in the AF + RHD group (12.08 per 100 person-years compared with 9.48 per 100 person-years for AF alone), though the difference was not statistically significant ($P = 0.54$; Figure 3). Similarly, stroke/TIA rates were similar between the 2 groups (HR: 0.97 [0.86, 1.11]; $P = 0.66$). Similar to the AF + HCM group, those with AF + RHD had comparable intracranial and gastrointestinal bleeding events with those with AF alone; however, other bleeding types were higher in patients with AF + RHD (HR: 1.24 [1.04, 1.49]; $P = 0.02$; Figure 3, Central Illustration).

AF + CA. All-cause mortality was higher, but not statistically significant, in the AF + CA group (event rate of 23.66 per 100 person-years compared with 9.48 per 100 person-years for AF alone), with HR: 1.47 (0.96, 2.26); $P = 0.08$. However, the AF + CA group was more likely to have stroke/TIA (HR: 2.00 [1.13, 3.54]; $P = 0.02$), gastrointestinal bleeding (HR: 2.50 [1.14, 5.47]; $P = 0.02$), and was overall associated with higher risk for the composite endpoint (HR: 1.63 [1.17, 2.27]; $P < 0.01$; Figure 3, Table 3).

SUBGROUP ANALYSIS. The subgroup analysis was performed according to sex, age, and race. Analysis demonstrated several important observations (Table 4, Figure 4).

On subgroup analysis, HR differed in patients with AF + SHD for certain outcomes compared with patients with AF only. The statistically significant outcomes were stroke/TIA (HR: 1.31 [1.06, 1.61], $P = 0.01$) in those <75 years, lower risk of stroke/TIA (HR: 0.85 [0.73, 0.99], $P = 0.04$) and higher risk of other bleeds (HR: 1.26 [1.03, 1.56], $P = 0.02$) in those ≥ 75 years, other bleeds (HR: 1.31 [1.06, 1.61], $P = 0.01$) in male

TABLE 1 Patient Characteristics

	AF only (n = 10,119)	AF + Cardiac Amyloidosis (n = 60)	AF + Hypertrophic Cardiomyopathy (n = 145)	AF + Rheumatic Heart Disease (n = 4,431)	Total (N = 14,755)	P Value
Age, years						<0.0001
Mean (SD)	76.1 (7.0)	77.7 (6.3)	74.7 (7.7)	77.5 (6.9)	76.5 (7.0)	
Gender						<0.0001
Female	4,280 (42.3)	19 (31.7)	66 (45.5)	2,089 (47.1)	6,454 (43.7)	
Male	5,839 (57.7)	41 (68.3)	79 (54.5)	2,342 (52.9)	8,301 (56.3)	
Race						0.6653
Non-White	1,931 (19.1)	11 (18.3)	27 (18.6)	884 (20.0)	2,853 (19.3)	
White	8,188 (80.9)	49 (81.7)	118 (81.4)	3,547 (80.0)	11,902 (80.7)	
Income range						0.0038
<\$40,000	2,834 (28.0)	12 (20.0)	34 (23.4)	1,346 (30.4)	4,226 (28.6)	
\$40,000-\$74,999	3,105 (30.7)	21 (35.0)	43 (29.7)	1,366 (30.8)	4,535 (30.7)	
\$75,000-\$124,999	2,689 (26.6)	15 (25.0)	41 (28.3)	1,143 (25.8)	3,888 (26.4)	
\$125,000-\$199,999	845 (8.4)	^a	^a	332 (7.5)	1,198 (8.1)	
\$200,000+	355 (3.5)	^a	^a	118 (2.7)	488 (3.3)	
Unknown	291 (2.9)	^a	^a	126 (2.8)	420 (2.8)	
Follow-up, years						0.8061
Median (IQR)	1.4 (0.8, 2.4)	1.4 (0.7, 2.3)	1.4 (0.8, 2.2)	1.4 (0.8, 2.4)	1.4 (0.8, 2.4)	
CHAD ₂ S ₂ -VASc						<0.0001
Mean (SD)	4.5 (1.4)	5.4 (1.5)	4.6 (1.4)	4.9 (1.4)	4.6 (1.4)	
HAS-BLED						<0.0001
Mean (SD)	2.9 (1.0)	3.1 (0.9)	2.9 (0.9)	3.1 (1.0)	3.0 (1.0)	
Comorbidities						
Chronic kidney disease	1,436 (14.2)	13 (21.7)	16 (11.0)	843 (19.0)	2,308 (15.6)	<0.0001
Heart failure	4,620 (45.7)	55 (91.7)	74 (51.0)	2,670 (60.3)	7,419 (50.3)	<0.0001
Hypertension	9,634 (95.2)	57 (95.0)	135 (93.1)	4,275 (96.5)	14,101 (95.6)	0.003
Thromboembolism	2,232 (22.1)	26 (43.3)	34 (23.4)	1,088 (24.6)	3,380 (22.9)	<0.0001
Vascular disease	6,709 (66.3)	39 (65.0)	106 (73.1)	3,249 (73.3)	10,103 (68.5)	<0.0001
Diabetes	4,201 (41.5)	25 (41.7)	48 (33.1)	1,813 (40.9)	6,087 (41.3)	0.2146
Anemia	5,921 (58.5)	40 (66.7)	85 (58.6)	2,966 (66.9)	9,012 (61.1)	<0.0001
Liver disease	1,183 (11.)	^a	^a	612 (13.8)	1,819 (12.3)	0.0032
Alcoholism	448 (4.4)	^a	^a	219 (4.9)	678 (4.6)	0.5373
Events in previous 12 months						
Heart failure hospitalization	519 (5.1)	12 (20.0)	13 (9.0)	487 (11.0)	1031 (7.0)	<0.0001
Stroke/systemic embolism	478 (4.7)	^a	^a	292 (6.6)	786 (5.3)	<0.0001
Transient ischemic attack	181 (1.8)	^a	^a	96 (2.2)	283 (1.9)	0.0008
Bleed	1958 (19.3)	16 (26.7)	37 (25.5)	1,085 (24.5)	3096 (21.0)	<0.0001
Gastrointestinal bleed	425 (4.2)	^a	^a	220 (5.0)	665 (4.5)	<0.0001
Intracranial bleed	1415 (14.0)	^a	^a	820 (18.5)	2264 (15.3)	<0.0001
Other bleed	146 (1.4)	^a	^a	62 (1.4)	213 (1.4)	0.5795
Events in previous 12 months						
Valve procedure	109 (1.1)	^a	^a	215 (4.9)	328 (2.2)	<0.0001
Medications filled in previous 90 days						
Antiplatelets	1097 (10.8)	^a	^a	607 (13.7)	1729 (11.7)	<0.0001
NSAIDs	691 (6.8)	^a	^a	257 (5.8)	964 (6.5)	0.0340
Warfarin	1010 (10.0)	^a	^a	500 (11.3)	1522 (10.3)	0.0134

Values are N(%), or median (IQR), or mean ± SD. ^aCell sizes less than 11 suppressed to protect patient confidentiality.
AF = atrial fibrillation; NSAIDs = nonsteroidal anti-inflammatory drugs.

patients, and other bleeds (HR: 1.24; 95% CI: 1.01-1.52; *P* = 0.04) in White patients.

FALSIFICATION POINTS. When examining the falsification points including (pneumonia, chronic obstructive pulmonary disease, fracture), the analysis

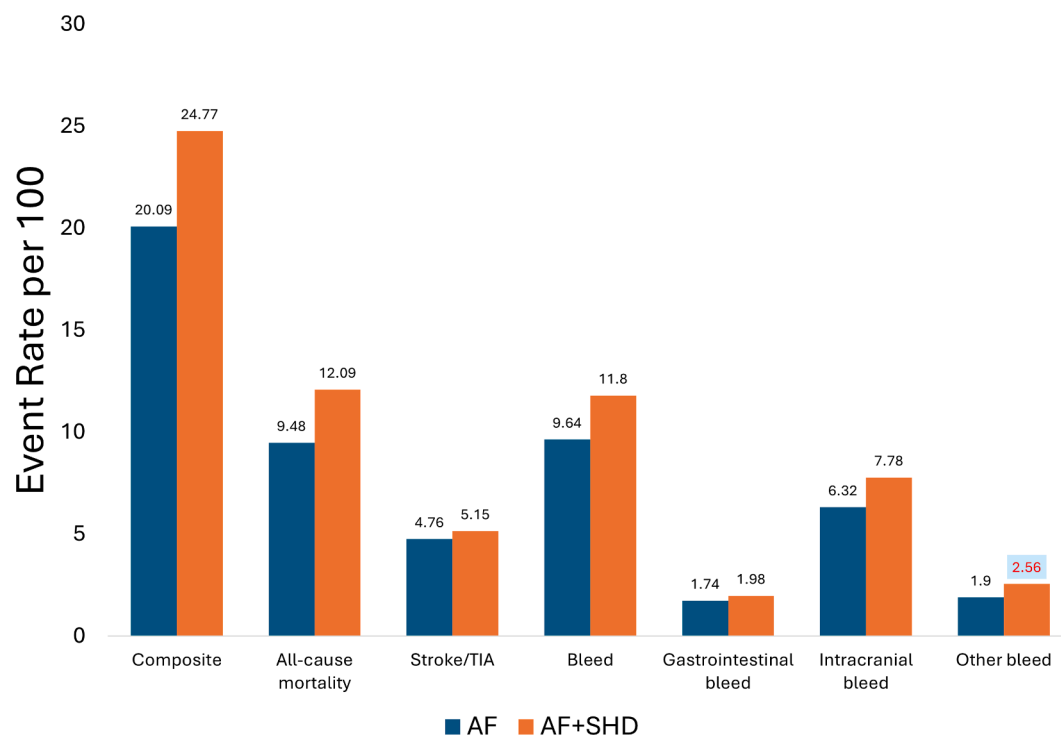
yielded no statistically significant results except for fracture in the AF + CA group (HR: 1.99 [1.07, 3.69]; *P* = 0.03; [Supplemental Figure 1](#)). Overall, this suggests that the observed findings are unlikely to be driven by confounding factors or spurious associations.

TABLE 2 Results of Cox Proportional Hazard Regression Comparing Post-LAAO Among Those With AF Only With Those With AF Plus Hypertrophic Cardiomyopathy, Rheumatic Heart Disease, and/or Cardiac Amyloidosis

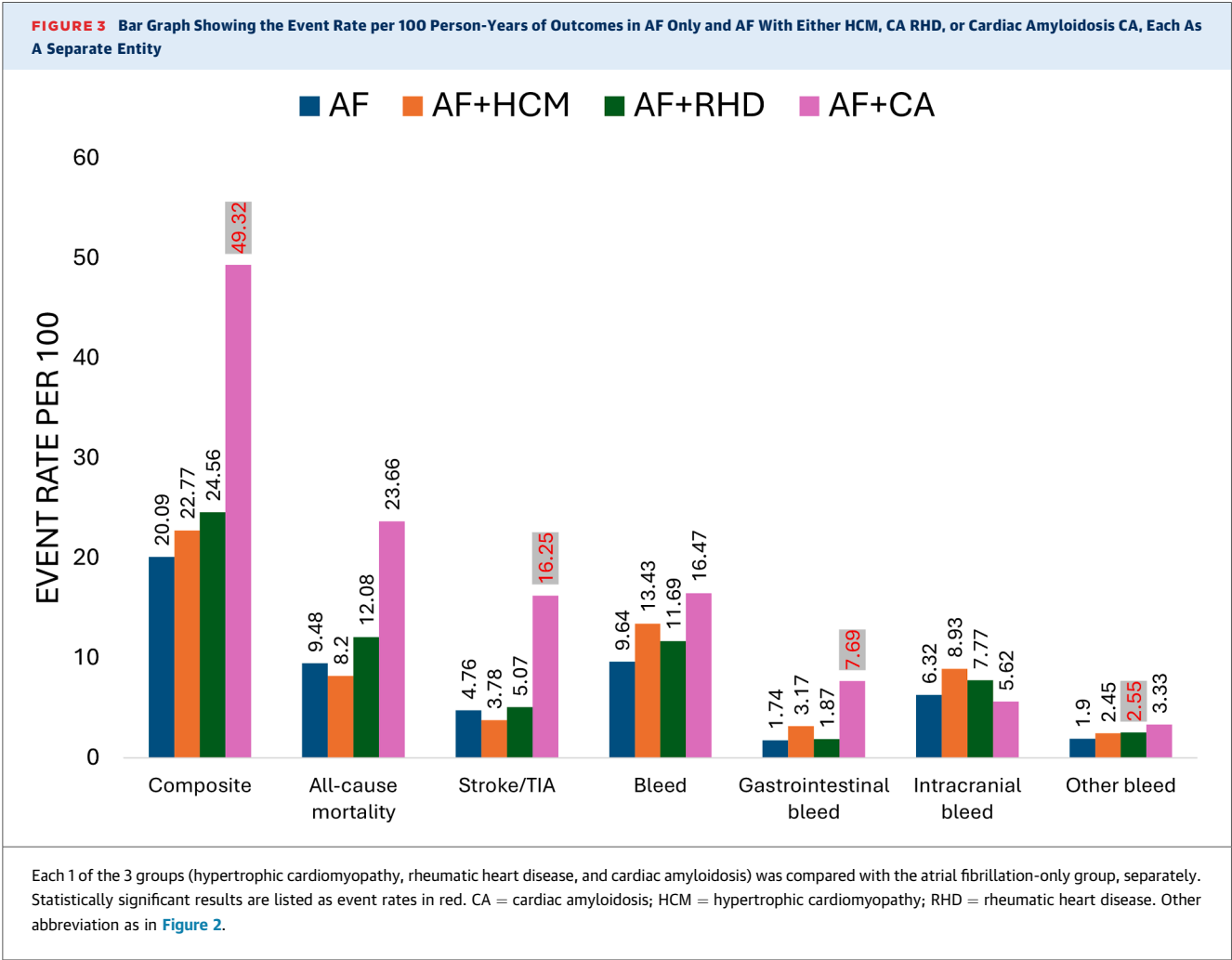
Outcomes	AF Only (ref)			AF + SHD			HR (95% CI)	P Value
	Number With Events	Person-Years	Event Rate per 100	Number With Events	Person- Years	Event Rate per 100		
	N = 10,119			N = 4,636				
Composite	3,080	15,332	20.09	1,703	6,875	24.77	1.03 (0.95, 1.12)	0.4560
All-cause mortality	1,716	18,100	9.48	1,010	8,356	12.09	1.04 (0.98, 1.10)	0.2241
Stroke/TIA	815	17,118	4.76	404	7,847	5.15	0.98 (0.87, 1.11)	0.7847
Bleed	1,547	16,049	9.64	853	7,229	11.80	1.04 (0.96, 1.14)	0.3380
Gastrointestinal bleed	310	17,790	1.74	162	8,186	1.98	1.05 (0.86, 1.27)	0.6653
Intracranial bleed	1,050	16,612	6.32	588	7,559	7.78	1.00 (0.90, 1.11)	0.9557
Other bleed	336	17,696	1.90	207	8,089	2.56	1.25 (1.05, 1.49)	0.0146
Falsification endpoints								
Pneumonia	1,157	16,937	6.83	593	7,732	7.67	0.94 (0.85, 1.05)	0.2615
COPD	787	17,147	4.59	411	7,804	5.27	0.89 (0.79, 1.01)	0.0670
Fracture	1,008	16,941	5.95	555	7,739	7.17	1.10 (1.00, 1.23)	0.0630

COPD = chronic obstructive pulmonary disease; LAOA = left atrial appendage occlusion; SHD = structural heart disease; TIA = transient ischemic attack. Other abbreviation as in Table 1.

FIGURE 2 Bar Graph Showing the Event Rate per 100 Person-Years of Outcomes in AF Only and AF With HCM, RHD, or CA, All Groups Combined



Statistically significant results are listed as event rate in red. AF = atrial fibrillation; AF-SHD = atrial fibrillation with hypertrophic cardiomyopathy, rheumatic heart disease, or cardiac amyloidosis; CA = cardiac amyloidosis; RHD = rheumatic heart disease; SHD = structural heart disease.



DISCUSSION

This study aimed to evaluate the difference in clinical outcomes following LAAO in patients with AF alone vs AF with HCM, RHD, or CA. Our study has several notable findings:

- Patients with CA had the highest event rates across all outcomes, indicative of a sicker population.
- Patients with AF and RHD had higher bleeding risk, whereas thromboembolic and mortality rates remained comparable with those with AF alone.
- Patients with AF and HCM had comparable clinical outcomes with those with AF only.

Previously, many studies have evaluated the challenges in managing patients with AF and HCM,^{12,13} RHD,^{14,15} or CA,^{1,16} often reporting overall worse outcomes in these populations. However, these investigations mainly focused on the general impact of AF on clinical outcomes. To the best of our

knowledge, this is the first study to evaluate the clinical outcomes after LAAO in these patient subgroups, thereby addressing an important gap in the literature.

COMPOSITE ENDPOINT AND MORTALITY OUTCOMES.

Patients with AF and underlying SHD (defined in this study as also having either HCM, RHD or CA) had a higher composite event rate (all-cause mortality, stroke, TIA, and bleeding) compared with those with AF alone. The absence of a significant difference in all-cause mortality suggests that, despite the varying pathophysiological profiles of these structural conditions, mortality risk in patients with AF may be influenced more by broader factors, underscoring the multifactorial complexity of mortality risk in AF populations. Although SHD contributes to clinical heterogeneity, it may not independently determine mortality outcomes in patients undergoing LAAO. However, it is crucial to

TABLE 3 Results Of Cox Proportional Hazard Regression Comparing Post-LAAO Events Among Those With AF Only With Those With AF Plus HCM, RHD, or CA

Hypertrophic Cardiomyopathy								
	AF only (ref)			AF + HCM			HR (95% CI)	P Value
	Number With Events	Person-Years	Event Rate per 100	Number With Events	Person- Years	Event Rate per 100		
Outcomes	N = 10,119			N = 145				
Composite	3,080	15,332	20.09	46	202	22.77	1.08 (0.81, 1.45)	0.5891
All-cause mortality	1,716	18,100	9.48	21	256	8.20	0.88 (0.57, 1.35)	0.5507
Stroke/TIA	815	17,118	4.76	^a	^a	3.78	0.82 (0.42, 1.61)	0.5700
Bleed	1,547	16,049	9.64	29	216	13.43	1.32 (0.91, 1.92)	0.1425
Gastrointestinal bleed	310	17,790	1.74	^a	^a	3.17	1.79 (0.85, 3.78)	0.1270
Intracranial bleed	1,050	16,612	6.32	20	224	8.93	1.31 (0.84, 2.06)	0.2372
Other bleed	336	17,696	1.90	^a	^a	2.45	1.31 (0.58, 2.95)	0.5133
Falsification endpoints								
Pneumonia	1,157	16,937	6.83	12	234	5.13	0.75 (0.43, 1.33)	0.3267
COPD	787	17,147	4.59	11	233	4.72	0.83 (0.46, 1.49)	0.5214
Fracture	1,008	16,941	5.95	12	233	5.15	0.91 (0.51, 1.62)	0.7374
Rheumatic Heart Disease								
	AF only (ref)			AF + RHD			HR (95% CI)	P Value
	Number With Events	Person-Years	Event Rate per 100	Number With Events	Person- Years	Event Rate per 100		
Outcomes	N = 10,119			N = 4,431				
Composite	3,080	15,332	20.09	1,621	6,599	24.56	1.03 (0.97, 1.09)	0.3895
All-cause mortality	1,716	18,100	9.48	967	8,008	12.08	1.03 (0.95, 1.11)	0.5409
Stroke/TIA	815	17,118	4.76	382	7,528	5.07	0.97 (0.86, 1.11)	0.6625
Bleed	1,547	16,049	9.64	810	6,927	11.69	1.03 (0.94, 1.13)	0.4947
Gastrointestinal bleed	310	17,790	1.74	147	7,843	1.87	0.99 (0.81, 1.22)	0.9327
Intracranial bleed	1,050	16,612	6.32	563	7,246	7.77	1.00 (0.90, 1.11)	0.9711
Other bleed	336	17,696	1.90	198	7,754	2.55	1.24 (1.04, 1.49)	0.0173
Falsification endpoints								
Pneumonia	1,157	16,937	6.83	571	7,409	7.71	0.95 (0.86, 1.05)	0.3142
COPD	787	17,147	4.59	395	7,481	5.28	0.90 (0.79, 1.01)	0.0812
Fracture	1,008	16,941	5.95	532	7,427	7.16	1.10 (0.99, 1.22)	0.0776
Cardiac Amyloidosis								
	AF only (ref)			AF + CA			HR (95% CI)	P Value
	Number With Events	Person-Years	Event Rate per 100	Number With Events	Person- Years	Event Rate per 100		
Outcomes	N = 10,119			N = 60				
Composite	3,080	15,332	20.09	36	73	49.32	1.63 (1.17, 2.27)	0.0043
All-cause mortality	1,716	18,100	9.48	22	93	23.66	1.47 (0.96, 2.26)	0.0796
Stroke/TIA	815	17,118	4.76	13	80	16.25	2.00 (1.13, 3.54)	0.0168
Bleed	1,547	16,049	9.64	14	85	16.47	1.27 (0.74, 2.18)	0.3915
Gastrointestinal bleed	310	17,790	1.74	^a	^a	7.69	2.50 (1.14, 5.47)	0.0221
Intracranial bleed	1,050	16,612	6.32	^a	^a	5.62	0.74 (0.30, 1.82)	0.5061
Other bleed	336	17,696	1.90	^a	^a	3.33	1.19 (0.37, 3.86)	0.7747
Falsification endpoints								
Pneumonia	1,157	16,937	6.83	^a	^a	11.24	1.07 (0.56, 2.05)	0.8309
COPD	787	17,147	4.59	^a	^a	5.56	0.88 (0.36, 2.15)	0.7744
Fracture	1,008	16,941	5.95	11	80	13.75	1.99 (1.07, 3.69)	0.0288

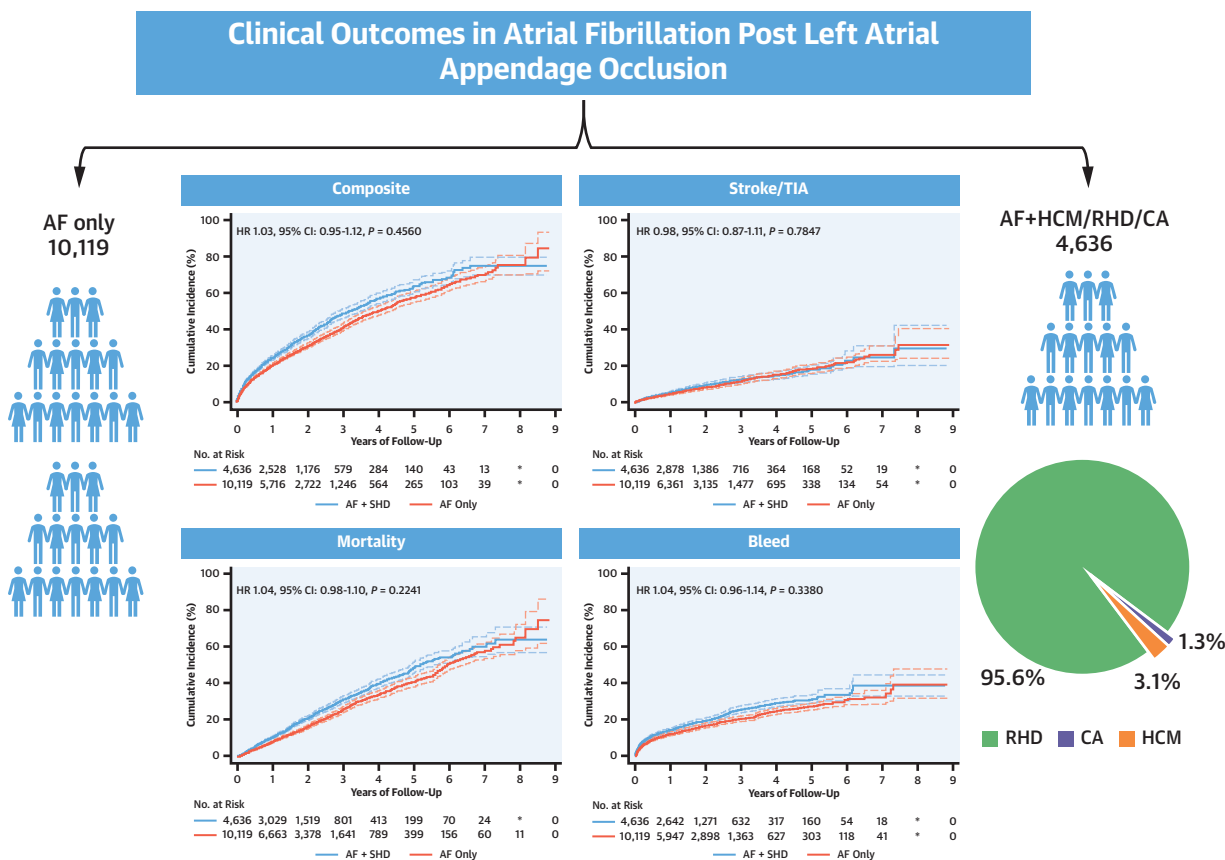
^aCell sizes less than 11 suppressed to protect patient confidentiality.

CA = cardiac amyloidosis; HCM = hypertrophic cardiomyopathy; RHD = rheumatic heart disease. Other abbreviations as in [Tables 1 and 2](#).

highlight that LAAO is designed to reduce the risk of stroke/TIA and bleeding. Thus, the absence of a demonstratable mortality benefit in this study should not be interpreted as diminishing its clinical value in appropriately selected high-risk patients.

HIGHER BLEEDING RISK IN PATIENTS WITH SHD. Our results demonstrate that patients with AF and CA or RHD had statistically significant increase in gastrointestinal and other bleeding, respectively. At baseline, groups had equivalent rate of oral

CENTRAL ILLUSTRATION Breakdown of the Patients Included in This Study Highlighting the Main Differences in Clinical Outcomes Between AF Only and AF With HCM, RHD, or CA, Post-Left Atrial Appendage Occlusion



Alabdjalbar MS, et al. JACC Clin Electrophysiol. 2026;12(5):1089-1101.

AF = atrial fibrillation; CA = cardiac amyloidosis; HCM = hypertrophic cardiomyopathy; RHD = rheumatic heart disease.

anticoagulants use; however, patients with AF + CA had a significantly higher rate of antiplatelet use. Post-LAAO, patients are typically maintained on anticoagulants/antiplatelets for a limited duration,¹⁷ which could put them at elevated risk of bleed. Thus, the agent(s) and duration of use post-LAAO are important factors to consider (ie, at 3 months, 16% of AF only were on anticoagulation as opposed to 30% of AF + CA, $P = 0.006$). Moreover, other internal or group-specific factors may have played a role. For instance, our results show that patients with AF + RHD were more likely to be on warfarin (before and after LAAO) than other groups, and those with AF + CA were more likely to be on anticoagulation at 3 months post-LAAO. Finally, patients with SHD, collectively, had higher mean scores of CHA₂DS₂-VASc (4.9 vs 4.5) and HAS-BLED (3.1 vs 2.9) than those with AF only

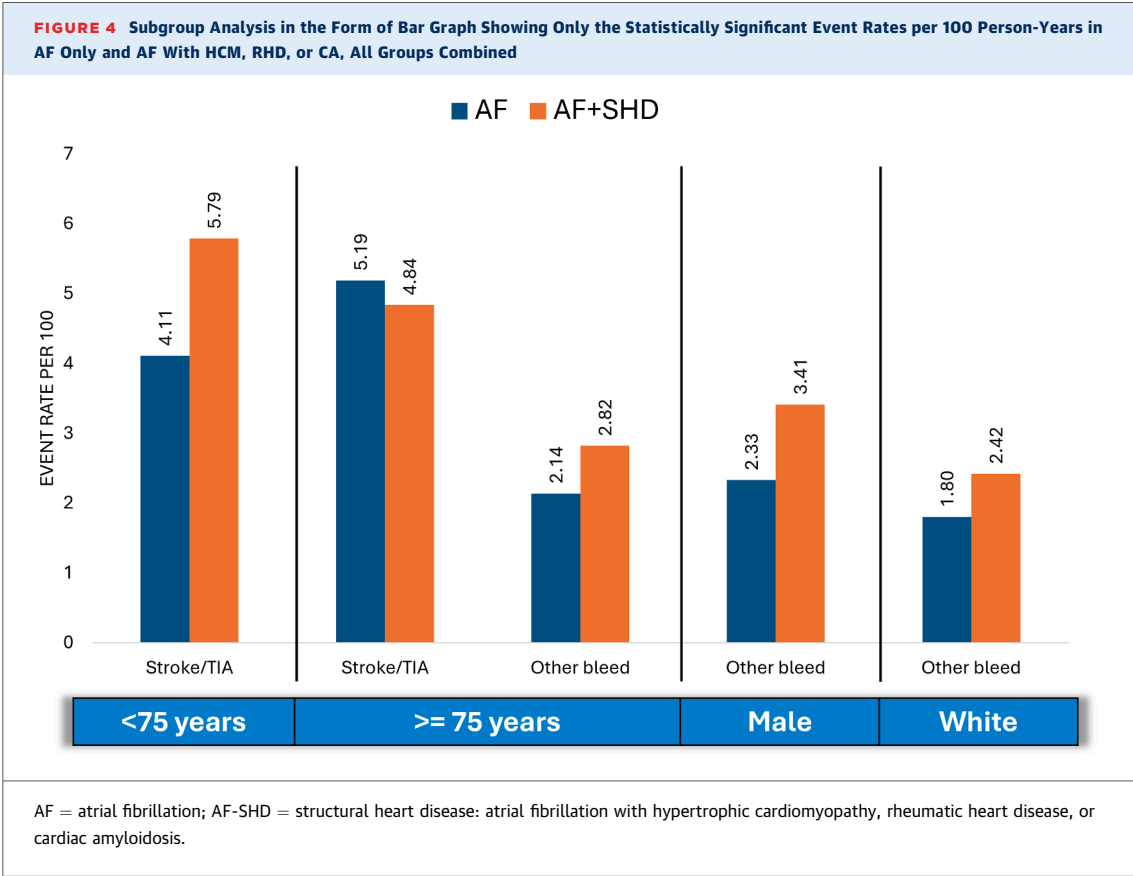
($P < 0.0001$), highlighting a sicker population. These findings underscore the need for personalized anticoagulation strategies tailored to the specific risks posed by underlying AF and SHD.

When comparing bleeding risk between our cohort with and without SHD and that of previous studies, the cohort included in this study was found to have similar¹⁸ (AF + HCM) or higher¹⁹ (AF + RHD) bleeding risk. As such, clinicians may need to adjust anticoagulation intensity, duration, or consider alternative therapeutic strategies for such high-risk individuals to mitigate the risk of hemorrhagic complications.

STROKE RATES NOT SIGNIFICANTLY ELEVATED DESPITE STRUCTURAL COMORBIDITY. Because of paucity of data regarding LAAO in pure cohorts of

TABLE 4 Subgroup Analysis of Cox Proportional Hazard Regression Comparing Post-LAAO Events Among Those With AF Only With Those With AF Plus HCM, RHD, or CA Based on Age, Gender, and Race

	AF only (ref)			AF + SHD			HR (95% CI)	P Value
	Number With Events	Person-Years	Event Rate per 100	Number With Events	Person- Years	Event Rate per 100		
Age								
Age < 75 years		N = 3,904			N = 1,444			
Composite	1,006	6,205	16.21	452	2,250	20.09	1.00 (0.89, 1.12)	0.9530
All-cause mortality	497	7,182	6.92	228	2,736	8.33	0.94 (0.80, 1.10)	0.4391
Stroke/TIA	281	6,832	4.11	147	2,541	5.79	1.31 (1.06, 1.61)	0.0118
Bleed	511	6,482	7.88	226	2,394	9.44	0.98 (0.83, 1.15)	0.7710
Gastrointestinal bleed	83	7,076	1.17	40	2,691	1.49	1.24 (0.83, 1.85)	0.2864
Intracranial bleed	372	6,659	5.59	159	2,481	6.41	0.89 (0.73, 1.07)	0.2185
Other bleed	108	7,060	1.53	54	2,663	2.03	1.22 (0.87, 1.71)	0.2415
Age 75+ years		N = 6,215			N = 3,192			
Composite	2,074	9,126	22.73	1,251	4,624	27.05	1.05 (0.98, 1.13)	0.1631
All-cause mortality	1,219	10,918	11.17	782	5,620	13.91	1.06 (0.97, 1.16)	0.2016
Stroke/TIA	534	10,294	5.19	257	5,306	4.84	0.85 (0.73, 0.99)	0.0367
Bleed	1,036	9,567	10.83	627	4,835	12.97	1.07 (0.97, 1.19)	0.1739
Gastrointestinal bleed	227	10,714	2.12	122	5,495	2.22	1.00 (0.79, 1.25)	0.9746
Intracranial bleed	678	9,952	6.81	429	5,078	8.45	1.06 (0.94, 1.20)	0.3558
Other bleed	228	10,636	2.14	153	5,426	2.82	1.26 (1.03, 1.56)	0.0284
Gender								
Male		N = 5,839			N = 2,462			
Composite	1,847	8,912	20.72	942	3,740	25.19	1.01 (0.93, 1.09)	0.8778
All-cause mortality	1,069	10,466	10.21	585	4,499	13.00	1.02 (0.92, 1.13)	0.7712
Stroke/TIA	454	9,907	4.58	201	4,262	4.72	0.94 (0.79, 1.12)	0.4989
Bleed	919	9,314	9.87	455	3,905	11.65	1.00 (0.89, 1.13)	0.9494
Gastrointestinal bleed	181	10,297	1.76	105	4,391	2.39	1.22 (0.94, 1.56)	0.1302
Intracranial bleed	592	9,676	6.12	267	4,134	6.46	0.87 (0.75, 1.01)	0.0636
Other bleed	237	10,180	2.33	147	4,315	3.41	1.31 (1.06, 1.61)	0.0124
Female		N = 4,280			N = 2,174			
Composite	1,233	6,419	19.21	761	3,135	24.27	1.07 (0.98, 1.18)	0.1387
All-cause mortality	647	7,633	8.48	425	3,857	11.02	1.05 (0.92, 1.19)	0.4702
Stroke/TIA	361	7,211	5.01	203	3,585	5.66	1.04 (0.87, 1.24)	0.6812
Bleed	628	6,735	9.32	398	3,324	11.97	1.08 (0.95, 1.23)	0.2450
Gastrointestinal bleed	129	7,493	1.72	57	3,794	1.50	0.79 (0.57, 1.09)	0.1471
Intracranial bleed	458	6,936	6.60	321	3,425	9.37	1.15 (1.00, 1.33)	0.0553
Other bleed	99	7,515	1.32	60	3,773	1.59	1.12 (0.80, 1.56)	0.5048
Race								
Non-White		N = 1,931			N = 922			
Composite	669	2,863	23.37	351	1,284	27.34	1.04 (0.91, 1.18)	0.6018
All-cause mortality	360	3,480	10.34	197	1,604	12.28	1.03 (0.86, 1.23)	0.7681
Stroke/TIA	179	3,275	5.47	98	1,496	6.55	1.03 (0.79, 1.34)	0.8159
Bleed	362	3,008	12.03	186	1,358	13.70	1.00 (0.83, 1.20)	0.9799
Gastrointestinal bleed	72	3,407	2.11	28	1,567	1.79	0.82 (0.51, 1.31)	0.4091
Intracranial bleed	252	3,131	8.05	136	1,423	9.56	0.99 (0.80, 1.22)	0.9016
Other bleed	78	3,377	2.31	49	1,547	3.17	1.29 (0.89, 1.86)	0.1763
White		N = 8,188			N = 3,714			
Composite	2,411	12,468	19.34	1,352	5,591	24.18	1.04 (0.97, 1.12)	0.2456
All-cause mortality	1,356	14,620	9.27	813	6,752	12.04	1.03 (0.94, 1.13)	0.5316
Stroke/TIA	636	13,842	4.59	306	6,351	4.82	0.96 (0.83, 1.10)	0.5577
Bleed	1,185	13,041	9.09	667	5,871	11.36	1.06 (0.96, 1.17)	0.2696
Gastrointestinal bleed	238	14,383	1.65	134	6,618	2.02	1.12 (0.90, 1.39)	0.3217
Intracranial bleed	798	13,481	5.92	452	6,135	7.37	1.01 (0.90, 1.14)	0.8684
Other bleed	258	14,319	1.80	158	6,542	2.42	1.24 (1.01, 1.52)	0.0403



HCM/RHD/CA, current guidelines focusing on these pathologies do not routinely recommend LAAO.¹³ Our data showed that stroke and TIA rates did not differ significantly between patients with AF and HCM and RHD groups, traditionally considered high risk regardless of their CHA₂DS₂-VASc scores and prone to thrombi outside the left atrial appendage. As such, this would support the use of LAAO in these patients, particularly when the bleeding risk is high. Further, the higher rates of stroke/TIA and mortality in the AF + CA group compared with AF alone may simply indicate a sicker patient population. However, data on the efficacy of LAAO vs anticoagulation in this high-risk population are lacking.

CLINICAL IMPLICATIONS FOR RISK STRATIFICATION AND ANTICOAGULATION DECISIONS. The present study highlights the importance of tailored anticoagulation strategies for patients with AF and coexisting SHD. The heterogeneous bleeding risks observed across subgroups suggest that conventional risk-stratification models may be insufficient in capturing the added complexity introduced by conditions such as HCM, RHD, and CA. Current anticoagulation guidelines, which primarily focus

on AF-related thromboembolic risk, should be refined to better address the additional bleeding risks posed by such conditions as HCM, RHD, and CA. Incorporating structural abnormalities, comorbidity burden, frailty, and functional status into decision-making frameworks could enhance risk-to-benefit assessments and help guide safer, more effective anticoagulation practices in this high-risk population while optimizing prevention of stroke.

STUDY LIMITATIONS AND NEED FOR PROSPECTIVE VALIDATION. Although our findings provide valuable insights, there are several limitations that should be acknowledged. The observational nature of this study shows associations but not causation. The small sample size in certain subgroups—particularly those with HCM and CA—limits the generalizability of our results and in some instances may have contributed to the lack of statistically significant results. The number of RHD patients is higher than what we expected. We used ICD-9/10 codes (supplemental table 1) to select RHD patients, but based on the numbers, there might be some patients who were incorrectly identified as RHD by their ICD codes. Yet it is the only

study that has evaluated these clinical outcomes combined. In addition, residual confounding factors (which were not available/obtainable), such as variations in anticoagulant types, renal function, frailty, and procedural outcomes (ie, peridevice leak, device type, operator experience, imaging follow-up), may influence our findings, as some of these factors may be directly related to some of the outcomes evaluated. However, it was reassuring to see the lack of significant associations in the majority of the falsification points, which reinforces the associations of the primary results. Further, the cross-sectional nature of our data prevents us from assessing long-term outcomes and may have been affected by issues related to databases (ie, measurement bias, coding) when pulling data. Because of the small sample size and multiple comparisons performed, the results in HCM and CA cohorts should serve as hypothesis generating. In addition, referral bias likely influenced the high bleeding rates across all cohorts. The OLDW was chosen over other registries because of its large nationally representative sample, real-world data across care settings, and comprehensive longitudinal claims that enable extended follow-up beyond the index hospitalization. Unlike procedure-focused registries, which may lack long-term tracking and are limited to specific cardiovascular interventions, OLDW supports broader outcome evaluation over time. Median follow-up durations for each group were included to further emphasize the longitudinal strengths and suitability of the dataset for the research question.

CONCLUSIONS

In this large retrospective analysis of patients undergoing LAAO, those patients with AF and either CA or RHD experienced higher bleeding risk compared

with patients with AF only. Notably, patients with HCM had outcomes similar to those with AF alone. The elevated risk in CA appears to reflect a sicker population overall, whereas bleeding in patients with RHD may relate to anticoagulation patterns. These findings suggest that LAAO remains a viable option for prevention of stroke, even in complex SHD populations. Prospective studies are needed to better inform individualized risk stratification and post-procedural management in these groups.

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PERSPECTIVES

COMPETENCY IN MEDICAL KNOWLEDGE:

Patients with atrial fibrillation and rheumatic heart disease or hypertrophic cardiomyopathy treated with left atrial appendage occlusion for prevention of stroke have similar outcomes to those with atrial fibrillation alone.

TRANSLATIONAL OUTLOOK: This study highlights the stable stroke risk in patients with atrial fibrillation with and without structural heart disease and calls for randomized trials to evaluate those outcomes post-left atrial appendage occlusion in a prospective manner.

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KEY WORDS atrial fibrillation, cardiac amyloidosis, hypertrophic cardiomyopathy, left atrial appendage occlusion, rheumatic heart disease

APPENDIX For supplemental figures and tables, please see the online version of this paper.



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